SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO2 ASSESSMENT TOOL 1 – Scenario-Based Requirement Analysis

Course Code:	CSA10 / CSA1063	**Course Title:**	Software Engineering
CO Assessed:	CO2	**Assessment:**	Assessment Tool 2 – Scenario-Based Requirement Analysis
Weightage:	20%	**Total Marks:**	25
CO2 Statement:	*Perform detailed requirements analysis, design scalable architectures, and prioritize features with a focus on cross-disciplinary skills*		
Student Name:	Brasilla R	**Reg. No.:**	192511357
Date:	23/10/2026	**Duration:**	60 minutes

Code of Conduct

I Brasilla R(192511357) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Brasilla R

Annexure A – Scenario-Based Requirement Analysis

Instructions to Students:
Each student will be assigned an individual software project problem statement. Analyze the given scenario and prepare a complete Software Requirement Analysis document by following the steps below.

Question

Based on the **assigned problem statement**, perform a **Scenario-Based Requirement Analysis** and prepare a Software Requirement Specification (SRS) document. Your analysis should include the following:

1. **Study and Analyze the Scenario**
   - Carefully understand the given problem statement.
   - Identify the business domain, stakeholders, existing challenges, and system objectives.
2. **Identify Functional and Non-Functional Requirements**
   - List all functional requirements required by the proposed system.
   - Identify suitable non-functional requirements such as performance, security, reliability, usability, scalability, maintainability, and availability.
3. **Identify Actors and Use Cases**
   - Identify all primary and secondary actors interacting with the system.
   - List the major use cases associated with each actor.
   - Draw a Use Case Diagram representing the interactions between actors and the system.
4. **Prepare the Software Requirement Specification (SRS)**
   Develop an SRS document containing:
   - Introduction
   - Problem Description
   - Scope of the System

- Functional Requirements
- Non-Functional Requirements
- Use Case Descriptions
- Data Requirements
- External Interface Requirements
- System Features

5. **Identify Assumptions and Constraints**
   - List the assumptions considered while developing the system.
   - Identify technical, operational, economic, legal, and time-related constraints affecting the project.
6. **Validate Requirement Completeness and Consistency**
   - Verify whether all stakeholder requirements have been addressed.
   - Check for ambiguity, redundancy, inconsistency, and missing requirements.
   - Suggest improvements to enhance the quality and completeness of the requirement specification.

Rubrics for Calculation

Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Scenario Understanding and Problem Analysis (30%)	Demonstrates thorough understanding of the scenario, stakeholders, objectives, and business context with clear analysis.	Demonstrates good understanding with minor omissions.	Basic understanding with limited analysis.	Poor understanding; major aspects missing or incorrect.
Functional and Non-Functional Requirement Identification (25%)	Clearly identifies comprehensive, accurate, and well-classified functional and non-functional requirements.	Identifies most requirements with minor classification errors.	Identifies only basic requirements; several omissions.	Requirements are incomplete, inaccurate, or poorly classified.
Actor Identification and Use Case Modelling (20%)	Correctly identifies all relevant actors and use cases; use case diagram is complete, accurate, and follows UML standards.	Most actors and use cases identified; diagram has minor errors.	Limited actors/use cases identified; diagram partially correct.	Actors/use cases missing or incorrect; diagram absent or inaccurate.
Software Requirement Specification (SRS)	SRS is well-structured, complete, professionally organized, and	SRS is organized with minor missing sections or formatting issues.	SRS includes only essential sections with limited detail.	SRS is incomplete, poorly organized, or

Documentation (15%)	includes all required sections.			lacks essential components.
Assumptions, Constraints, and Requirement Validation (10%)	Clearly identifies realistic assumptions and constraints; validates requirements for completeness, consistency, and ambiguity with appropriate justification.	Identifies most assumptions and constraints; validation is adequate.	Limited assumptions/constraints; validation is superficial.	Assumptions, constraints, and validation are missing or incorrect.

Criteria	**Mark**
Scenario Understanding and Problem Analysis (30%)	/5
Functional and Non-Functional Requirement Identification (25%)	/5
Actor Identification and Use Case Modelling (20%)	/5
Software Requirement Specification (SRS) Documentation (15%)	/5
Assumptions, Constraints, and Requirement Validation (10%)	/5
TOTAL	/25

Software Requirement Specification (SRS)

Vehicle Service and Maintenance Management

PROBLEM STATEMENT:

Vehicle Service and Maintenance Management System: Develop a system that allows vehicle owners to book service appointments, maintain service history, receive maintenance reminders, and track service status.

1. Study and Analysis of the Scenario

1.1 Problem Statement

Develop a system that allows vehicle owners to book service appointments, maintain service history, receive maintenance reminders, and track service status.

1.2 Business Domain

The system belongs to the automobile service and maintenance domain. It supports vehicle service centres, garages and workshops that service cars, two-wheelers and commercial vehicles, and the customers who own these vehicles. It combines appointment scheduling, job management, customer relationship management and notification services in a single web and mobile based platform.

1.3 Stakeholders

- **Vehicle Owner (Customer):** Registers vehicles, books appointments, tracks progress, views history, pays bills and receives reminders.
- **Service Advisor / Receptionist:** Confirms appointments, prepares job cards and estimates, and communicates with customers.
- **Mechanic / Technician:** Performs the service and updates work progress and parts used.
- **System Administrator:** Manages users, service packages, pricing, branches and system configuration.
- **Service Centre Owner / Manager:** Monitors workload, revenue, customer satisfaction and performance reports.
- **Payment Gateway and SMS / Email Service (external systems):** Provide online payment processing and notification delivery.

1.4 Existing Challenges

- Customers must call or visit the workshop to book a service, which leads to long waiting times and double bookings.
- Service records are kept on paper or in disconnected spreadsheets, so they are easily lost and hard to retrieve.
- Owners forget service due dates or mileage limits, causing breakdowns, reduced vehicle life and lost repeat business.

- Customers have no visibility of the progress of their vehicle and must repeatedly call the workshop for updates.
- Workshop staff cannot balance workload, bay availability and technician allocation efficiently.
- Billing disputes arise because estimates, parts used and work done are not recorded transparently.

1.5 System Objectives

- To provide an online facility for booking, rescheduling and cancelling service appointments.
- To maintain a complete, secure and searchable digital service history for every registered vehicle.
- To send automatic maintenance reminders based on date and odometer reading.
- To let owners track the real-time status of their vehicle from check-in to delivery.
- To improve workshop efficiency, transparency in billing and overall customer satisfaction.

2. Functional and Non-Functional Requirements

2.1 Functional Requirements

FR1. The system shall allow vehicle owners to register, log in and manage their profile using email or mobile number with OTP verification.

FR2. The system shall allow an owner to add, edit and delete multiple vehicles with details such as registration number, make, model, year, fuel type and current odometer reading.

FR3. The system shall display the available service packages, estimated prices, branches and open time slots.

FR4. The system shall allow an owner to book a service appointment by selecting vehicle, service type, branch, date and time slot.

FR5. The system shall prevent double booking by allowing a slot to be booked only while workshop capacity is available.

FR6. The system shall allow an owner to reschedule or cancel an appointment before service begins and shall record the reason for cancellation.

FR7. The system shall send booking confirmation, rescheduling and cancellation notices through SMS, email and in-app notification.

FR8. The service advisor shall be able to view, approve, reject or reassign appointments and assign a mechanic to each job.

FR9. The system shall create a job card when the vehicle is checked in, including customer complaints, existing damage and an estimate for approval by the owner.

FR10. The mechanic shall be able to update the service status (Booked, Vehicle Received, Inspection, Work in Progress, Waiting for Parts, Quality Check, Completed, Delivered) with time stamps.

FR11. The system shall let the owner track the current service status and the expected completion time in real time and notify the owner on every status change.

FR12. The system shall record service details such as work done, parts replaced, labour charges, mechanic remarks, odometer reading and service date.

FR13. The system shall maintain a complete service history for every vehicle and allow the owner to view, search, filter and download it as a PDF.

FR14. The system shall calculate the next service due date and odometer value based on the service type and manufacturer schedule.

FR15. The system shall automatically send maintenance reminders before the due date or mileage, and a follow-up reminder if the service is not booked.

FR16. The system shall let the owner set reminder preferences such as channel, advance days and snooze.

FR17. The system shall generate itemised invoices and support online payment, cash payment and payment receipts.

FR18. The system shall allow owners to submit ratings and feedback after service completion.

FR19. The administrator shall be able to manage users, roles, service packages, prices, branches, working hours and spare parts.

FR20. The system shall generate reports on appointments, revenue, technician workload, pending services and customer feedback.

FR21. The system shall maintain an audit log of important actions such as status updates, cancellations and changes to records.

2.2 Non-Functional Requirements

- **Performance:** Pages shall load within 3 seconds, and booking and status queries shall respond within 2 seconds for up to 1,000 concurrent users.
- **Security:** All passwords shall be hashed, data shall be transmitted over HTTPS (TLS), role-based access control shall be enforced, and sessions shall expire after inactivity. Payment data shall follow PCI-DSS guidelines.
- **Reliability:** The system shall be capable of handling failures gracefully, with no data loss on failure and automatic retry of failed notifications.
- **Availability:** The system shall be available 99.5% of the time, with scheduled maintenance outside working hours and daily backups.
- **Usability:** The interface shall be simple, responsive on mobile and desktop, require no training for owners, and support English and regional languages such as Tamil.
- **Scalability:** The architecture shall support addition of new branches, vehicles and users without redesign, using a modular and horizontally scalable design.
- **Maintainability:** The system shall use modular, well-documented code, so that changes and bug fixes can be made with minimal impact on other modules.
- **Portability and Compatibility:** The system shall work on current versions of Chrome, Edge, Firefox and Safari and on Android and iOS devices.
- **Data Integrity and Privacy:** Customer data shall be protected as per applicable data protection laws, and service history shall not be modified without an audit trail.

3. Actors and Use Cases

3.1 Primary Actors

- **Vehicle Owner:** Register / Login, Manage Vehicle Profile, Book Service Appointment, Reschedule / Cancel Appointment, Track Service Status, View Service History, Receive Maintenance Reminders, Make Payment, Submit Feedback.
- **Service Advisor:** Manage Appointments, Create Job Card and Estimate, Schedule Maintenance Reminders, Generate Invoice.
- **Mechanic / Technician:** Update Service Status, Record Service Details.
- **System Administrator:** Register / Login, Manage Users and Service Catalogue, Generate Reports.

3.2 Secondary Actors

- **Payment Gateway:** Processes online payments for Make Payment and Generate Invoice.
- **SMS / Email Service:** Delivers reminders, confirmations and status alerts for Receive Maintenance Reminders and Update Service Status.

3.3 Use Case Diagram

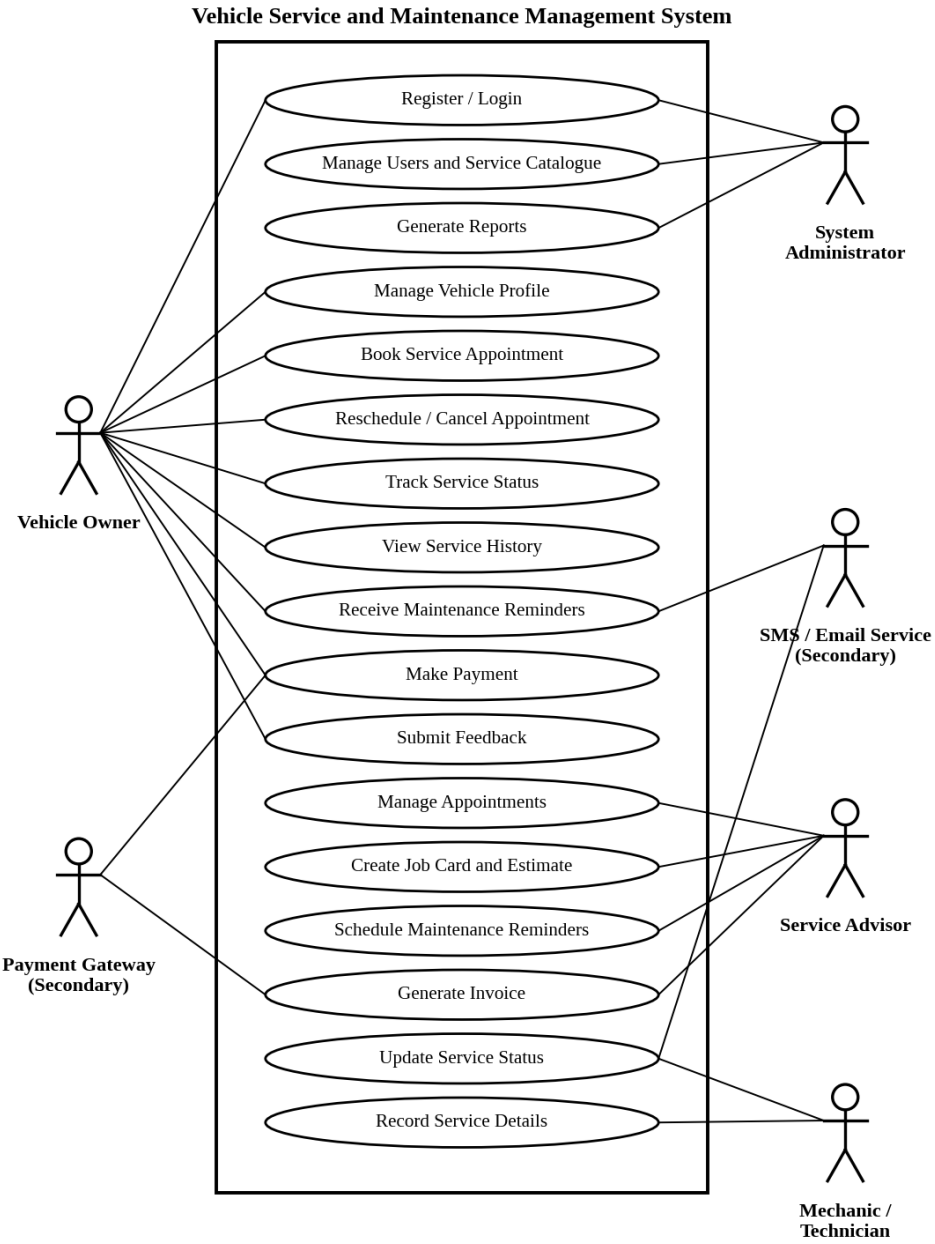


Figure 1: Use Case Diagram of the Vehicle Service and Maintenance Management System

4. Software Requirement Specification (SRS)

4.1 Introduction

Purpose: This SRS describes the requirements of the Vehicle Service and Maintenance Management System (VSMMS). It is intended for developers, testers, project managers and the service centre management as the basis for design, development and acceptance.

Intended Audience: Software development team, quality assurance team, service centre management and end users.

Definitions: SRS - Software Requirement Specification; VSMMS - Vehicle Service and Maintenance Management System; OTP - One Time Password; Job Card - a record of the work to be done on a vehicle.

4.2 Problem Description

Vehicle service centres still depend on phone calls, walk-ins and paper records. This results in missed appointments, lost service history, poor communication of progress and customers who forget scheduled maintenance. The proposed system replaces these manual practices with an online platform on which owners can book services, view their complete service history, receive timely reminders and follow the progress of their vehicle, while the service centre manages its schedule, jobs and billing efficiently.

4.3 Scope of the System

In scope: User and vehicle management, appointment booking, job card and service status tracking, service history, maintenance reminders, invoicing and online payment, feedback, administration and reports.

Out of scope: Spare parts procurement and supplier management, payroll and accounting of the workshop, vehicle insurance processing and roadside assistance.

4.4 Functional Requirements

The functional requirements FR1 to FR21 listed in Section 2.1 form part of this SRS and are grouped as follows: User and Vehicle Management (FR1-FR2), Appointment Management (FR3-FR8), Service Execution and Status Tracking (FR9-FR12), Service History (FR13), Maintenance Reminders (FR14-FR16), Billing and Feedback (FR17-FR18), and Administration and Reporting (FR19-FR21).

4.5 Non-Functional Requirements

The non-functional requirements on performance, security, reliability, availability, usability, scalability, maintainability, portability and data privacy listed in Section 2.2 form part of this SRS and shall be considered mandatory design constraints.

4.6 Use Case Descriptions

Use Case 1: Book Service Appointment

Primary Actor: Vehicle Owner

Precondition: The owner is logged in and has at least one registered vehicle.

Main Flow:

1. The owner selects the option Book Service.

2. The system displays the registered vehicles and the owner selects one.
3. The owner chooses the service type, branch and preferred date.
4. The system displays the available time slots.
5. The owner selects a slot and confirms the booking.
6. The system saves the appointment, generates a booking ID and sends a confirmation by SMS and email.

Alternate Flow: If the chosen slot is already full, the system shows the nearest available slots. If the owner cancels the process, no booking is created.

Postcondition: The appointment is stored with the status Booked and the service advisor can view it.

Use Case 2: Track Service Status

Primary Actor: Vehicle Owner

Precondition: The vehicle has been booked or checked in at the service centre.

Main Flow:

1. The owner opens the Track Service page and selects the booking.
2. The system retrieves the latest status and time stamps from the job card.
3. The system displays the current stage, work completed, estimated completion time and any pending approvals.
4. When the mechanic updates the status, the system sends a notification to the owner.

Alternate Flow: If the estimate has been revised, the owner is asked to approve or reject it before work continues.

Postcondition: The owner is aware of the latest status of the vehicle at any time.

Use Case 3: View Service History

Primary Actor: Vehicle Owner

Precondition: The owner is logged in and the vehicle has at least one completed service.

Main Flow:

1. The owner selects a vehicle and opens Service History.
2. The system lists all past services in date order, with the work done, parts replaced, cost and odometer reading.
3. The owner searches or filters the records by date or service type.
4. The owner opens a record to view the invoice or downloads the history as a PDF.

Alternate Flow: If no service has been recorded, the system displays the message that no history is available.

Postcondition: The owner has an accurate record of all services of the vehicle.

Use Case 4: Receive Maintenance Reminders

Primary Actor: SMS / Email Service (secondary) and Vehicle Owner

Precondition: The vehicle has a recorded last service date and odometer reading, and reminders are enabled.

Main Flow:

1. The system calculates the next service due date and mileage after every completed service.
2. A scheduled job checks daily for vehicles whose due date is within the reminder period.
3. The system prepares the reminder with vehicle details and a link to book the service.
4. The notification service sends the reminder through the owner's selected channel.
5. The system records the delivery status.

Alternate Flow: If delivery fails, the system retries and then sends it by an alternate channel. If the owner has already booked, the reminder is cancelled.

Postcondition: The owner is reminded in time and can book the service directly.

Use Case 5: Update Service Status

Primary Actor: Mechanic / Technician

Precondition: A job card has been created and assigned to the mechanic.

Main Flow:

1. The mechanic opens the assigned job card.
2. The mechanic selects the new status and enters remarks, parts used and the work completed.
3. The system saves the update with a time stamp and the mechanic's ID.
4. The system notifies the owner and the service advisor.

Alternate Flow: If extra work is required, the mechanic raises a request and the advisor obtains the owner's approval before continuing.

Postcondition: The job card is updated and the owner can see the new status.

4.7 Data Requirements

The system shall store the following main data entities. Each entity is identified by a unique ID and related to the others as described.

- **User:** user ID, name, email, mobile number, password hash, role, address, status.
- **Vehicle:** vehicle ID, owner ID, registration number, make, model, year, fuel type, odometer reading.
- **Service Package:** package ID, name, description, price, recommended interval in months and kilometres.
- **Appointment:** appointment ID, vehicle ID, package ID, branch ID, date, time slot, status, created time.
- **Job Card:** job card ID, appointment ID, mechanic ID, complaints, estimate, current status, start time, end time.
- **Service Record:** record ID, job card ID, work done, parts replaced, labour cost, odometer reading, service date.
- **Reminder:** reminder ID, vehicle ID, due date, due mileage, channel, status, sent time.
- **Invoice and Payment:** invoice ID, job card ID, itemised charges, tax, total, payment mode, transaction ID, payment status.

- **Feedback:** feedback ID, job card ID, rating, comments.
- **Branch and Audit Log:** branch details and working hours; log of user, action and time.

Data shall be stored in a relational database with referential integrity, backed up daily, and service records shall be retained for at least five years.

4.8 External Interface Requirements

- **User Interface:** A responsive web application and mobile application with dashboards for owner, advisor, mechanic and administrator, using forms, calendars and status timelines.
- **Hardware Interface:** Standard devices such as desktops, smartphones and tablets with an internet connection; an optional barcode or QR scanner at check-in.
- **Software Interface:** A relational database (MySQL / PostgreSQL), a REST API between the front end and back end, a payment gateway API, an SMS gateway API and an email (SMTP) service.
- **Communication Interface:** HTTPS for all communication, JSON for data exchange, and push notifications for mobile devices.

4.9 System Features

- **Appointment Booking:** Online booking, rescheduling and cancellation with real-time slot availability (FR3-FR8).
- **Service Status Tracking:** Live job progress with notifications at every stage (FR9-FR11).
- **Service History Management:** Digital, searchable history of every service for each vehicle (FR12-FR13).
- **Maintenance Reminders:** Automated reminders based on date and mileage (FR14-FR16).
- **Billing and Payments:** Itemised invoices and secure online payment (FR17).
- **Administration and Reporting:** Management of users, packages and branches, with reports and audit logs (FR19-FR21).

5. Assumptions and Constraints

5.1 Assumptions

- Vehicle owners have a smartphone or computer with internet access and a valid mobile number or email address.
- Owners enter the odometer reading and vehicle details correctly.
- The service centre staff are trained to update the job status regularly.
- Third-party payment, SMS and email services are available and reliable.
- Manufacturer service schedules are available for calculating due dates.
- Each vehicle is owned by a single registered owner at a given time.

5.2 Constraints

- **Technical:** Dependence on internet connectivity and on third-party APIs; must run on the common browsers and mobile platforms.

- **Operational:** Workshop capacity such as bays and technicians limits the number of slots; staff must update status in real time for tracking to be accurate.
- **Economic:** Limited project budget, along with recurring costs of hosting, SMS and payment gateway charges.
- **Legal:** Compliance with data protection and privacy laws, the IT Act, and payment security standards; customer consent is needed for sending reminders.
- **Time:** The project has to be completed within the academic or project schedule, so the features are delivered in phases, with core booking, tracking, history and reminders first.

6. Validation of Requirement Completeness and Consistency

6.1 Completeness

Every requirement in the problem statement has been mapped to at least one functional requirement and use case: booking of service appointments (FR3-FR8, Use Case 1), maintaining service history (FR12-FR13, Use Case 3), maintenance reminders (FR14-FR16, Use Case 4) and tracking service status (FR9-FR11, Use Case 2). The needs of all stakeholders, namely owner, advisor, mechanic, administrator and management, are covered through their functions, reports and non-functional requirements.

6.2 Consistency, Ambiguity and Redundancy

- Consistency: The status values in FR10 are used uniformly in the use cases, the job card entity and the notifications, and no two requirements contradict each other.
- Ambiguity: Terms such as quickly and real time were replaced with measurable values (for example, response within 2 seconds, availability of 99.5%).
- Redundancy: Common functions such as notification and authentication are specified once and reused by all the use cases.
- Missing requirements checked: Cancellation rules, payment failure handling, notification failure and audit trail were identified and added.

6.3 Suggested Improvements

- Define clear policies for cancellation time limits and rescheduling charges.
- Add support for pick-up and drop-off of vehicles and a service-on-wheels option.
- Include integration with spare parts inventory so that availability affects the expected completion time.
- Provide digital document storage for the vehicle's insurance, pollution certificate and warranty, with expiry reminders.
- Prepare acceptance criteria for each requirement and review the SRS with stakeholders through walkthroughs and prototypes.
- Use a requirements traceability matrix maintained during design and testing, so that changes can be managed.

Conclusion

The SRS given above analyses the scenario, defines the functional and non-functional requirements, identifies the actors and use cases, and records the assumptions, constraints and validation of the Vehicle Service and Maintenance Management System. It provides a clear and complete basis for the design and development of the system.